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Introduction to Data Science and Systems

Probabilities II – Bayes' Rule

Dr Nicolas Pugeault

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Today's learning outcomes

- conditional, marginal and joint distribution
- what entropy is and how it can be computed
- Bayes' rule and how to apply it problems
- what bigrams are and how they can be used



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Part 1 - Joint, marginal and conditional probabilities





Last week: Romeo & Juliet in letters



The code in the lecture notes loads a text file (in this case, Romeo and Juliet), and converts into a vector of numerical codes. It keeps only letters and spaces and converts everything to lowercase.

*Rom. Give me a torch. I am not for this ambling.
Being but heavy, I will bear the light.*

Mer. Nay, gentle Romeo, we must have you dance.

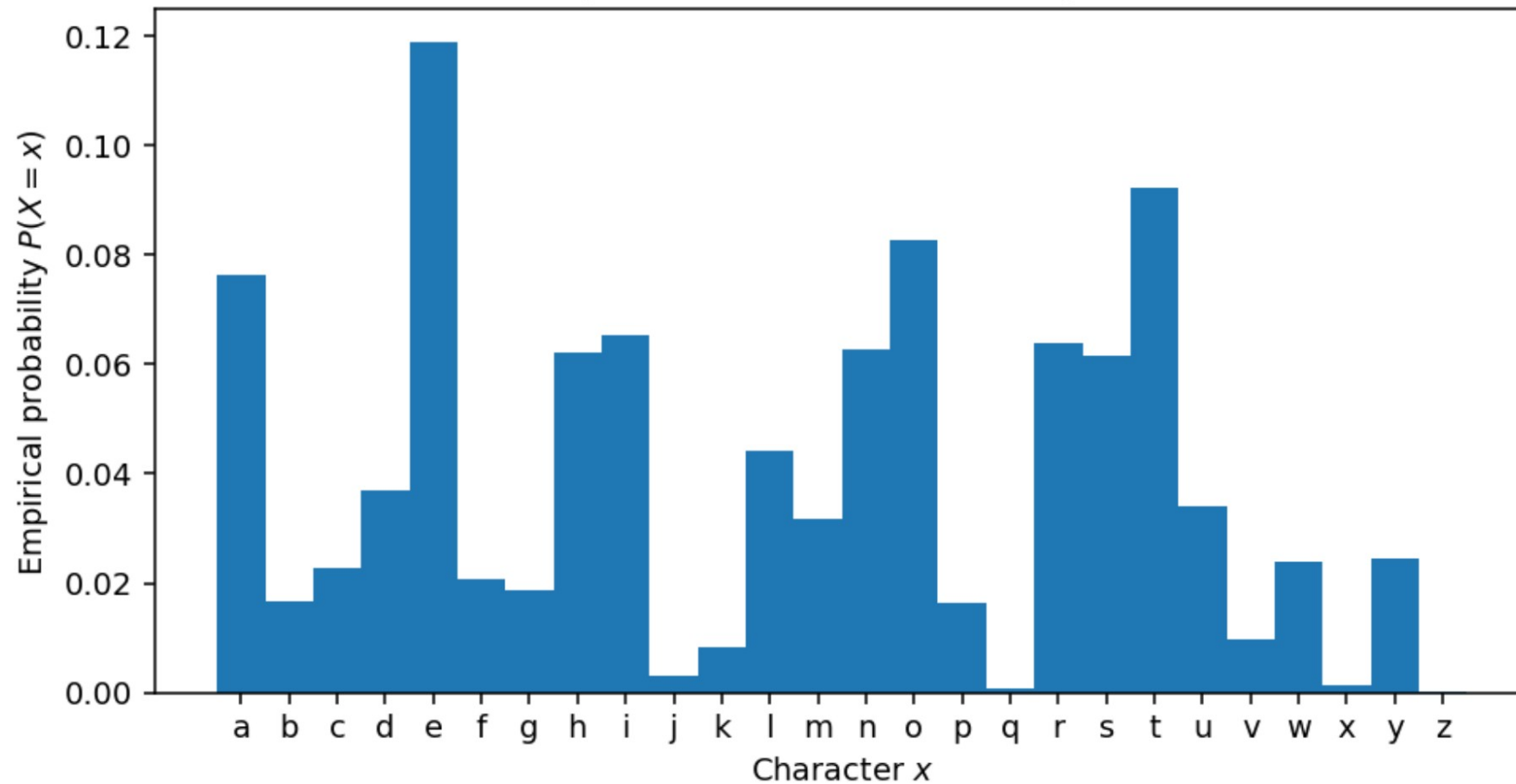
*Rom. Not I, believe me. You have dancing shoes
With nimble soles; I have a soul of lead
So stakes me to the ground I cannot move.*



Letters PMF in Romeo & Juliet



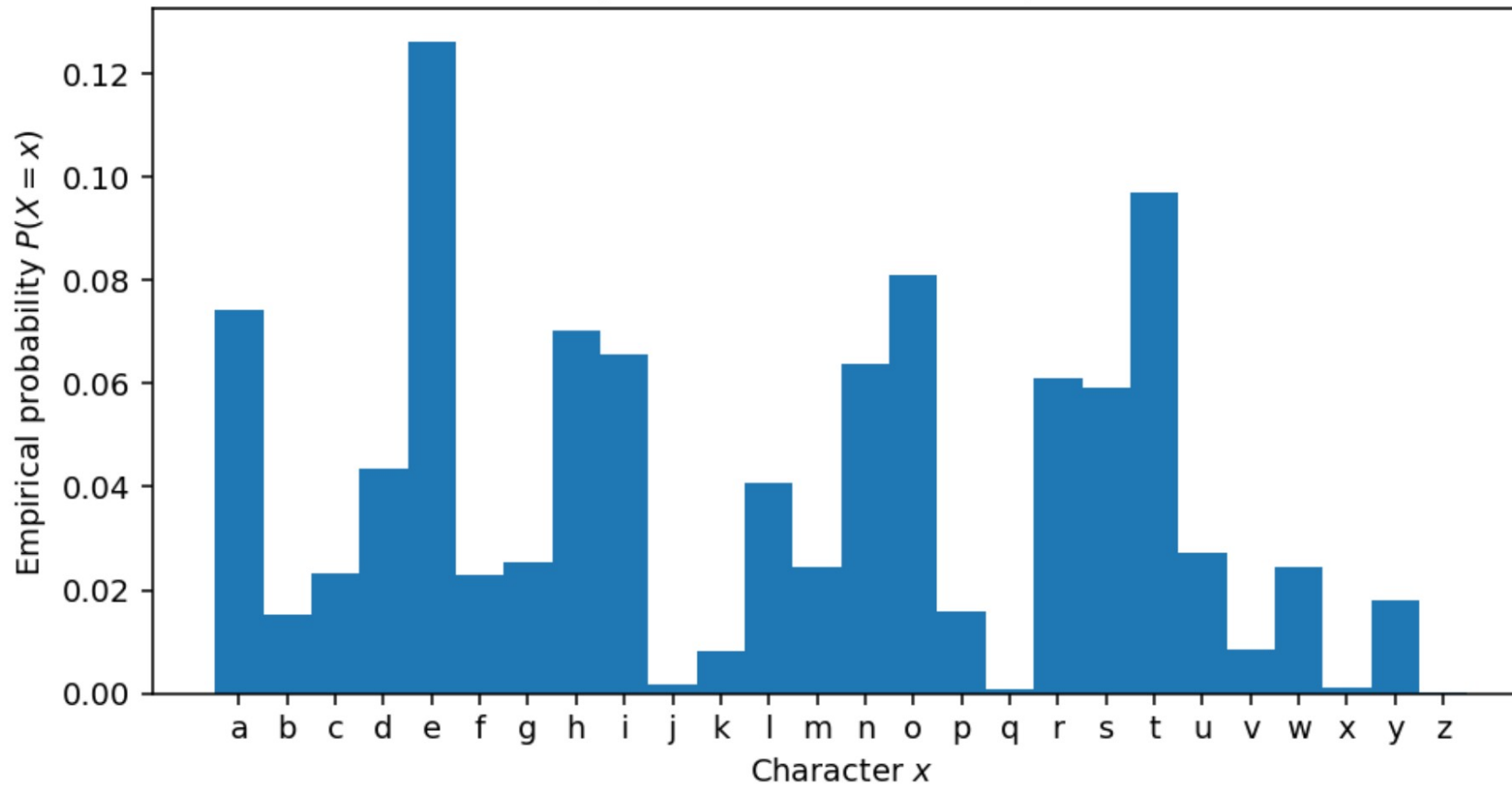
Empirical PMF of Romeo and Juliet





Letters PMF in Metamorphosis

Empirical PMF of Metamorphosis



Text generation from letter distribution

The results are similar (both are English texts), but there are some subtle differences; there slightly more q characters in Kafka compared to Shakespeare; the frequencies of h and i are reversed.

As always, given a PMF, we can draw samples from it with our standard procedure, though the results aren't very exciting:



https://en.wikipedia.org/wiki/Infinite_monkey_theorem

```
[60]: dict_pmf = {outcome:p for outcome,p in zip(alphabet, rj_pmf)}  
      """.join(discrete_sample(dict_pmf, 140))
```

```
[60]: 'flfsuagsgdhgumierotoaecnoobhftlrkxbudmrgslseadmtiigmwhnmeteetbcarihgenietesiioasktorntmpoeahamnyrhfhgqteirltspamyeeitaagiaeabntnethbslbbam'
```

Likelihood function

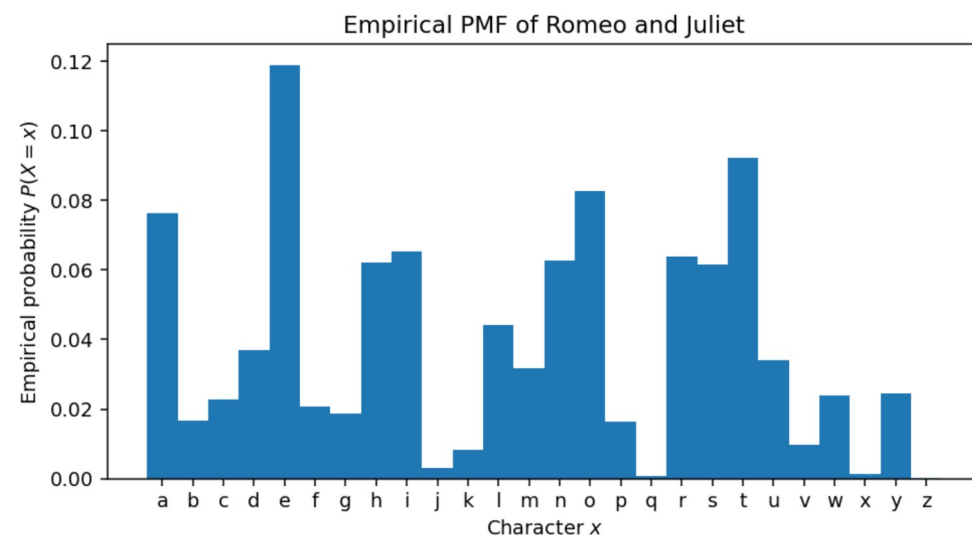
- When talking about **likelihood**, we often write $L(x_i)$ to mean the likelihood of x_i . The likelihood is not a probability. It is a function of data, and $L(x_i) = f_x(x_i)$
- For example, consider the empirical PMF of Romeo and Juliet. This gives the probability of seeing any given character. The likelihood of seeing all of the characters, given our per-character probability model, is:

$$\mathcal{L}(c_1, c_2, \dots, c_n) = \prod_i P(c_i)$$

*Rom. Give me a torch. I am not for this ambling.
Being but heavy, I will bear the light.*

Mer. Nay, gentle Romeo, we must have you dance.

*Rom. Not I, believe me. You have dancing shoes
With nimble soles; I have a soul of lead
So stakes me to the ground I cannot move.*





Romeo & Juliet: Text likelihood with underflow error!

This is the likelihood of the text.

```
[71]: def likelihood(sequence, p):  
  
    result = 1.0  
    for char in sequence:  
        # this is bad: multiplying lots of small numbers  
        # will underflow very quickly  
        result = result * p[char]  
    return result
```

```
[72]: print(likelihood(rj_chars, rj_pmf)) # huh?
```

```
0.0
```



Romeo & Juliet: Text log-likelihood

The log-likelihood does not have this problem with underflow:

$$\log \mathcal{L}(c_1, c_2, \dots, c_n) = \sum_i \log P(c_i)$$

```
[74]: def log_likelihood(sequence, p):  
    result = 0.0  
    for char in sequence:  
        ## summing logs is stable numerically  
        result = result + np.log(p[char])  
    return result
```

```
[75]: #  
print(log_likelihood(rj_chars, rj_pmf))
```

```
-344778.03999527503
```



Who wrote Romeo & Juliet: A statistical approach

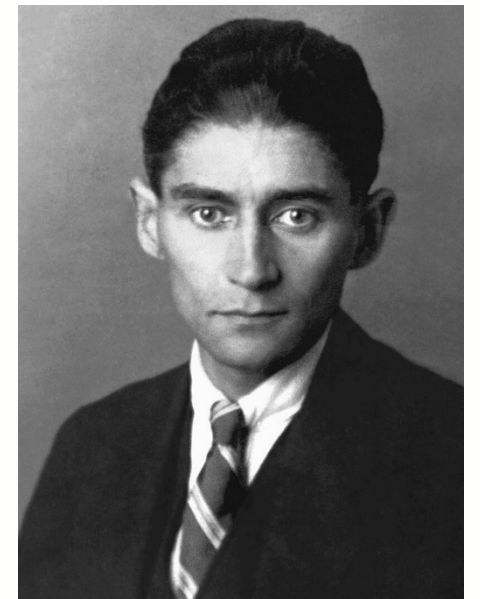
We could imagine that writing plays and novels is an activity that mysterious entities do by generating random characters according to a PMF.

Given an "unknown" text (in this case *Macbeth*), how likely is it that:

- A. It was generated by a mysterious entity using the PMF for *Romeo and Juliet*
- B. It was generated by a mysterious entity using the PMF for *Metamorphosis*

Neither of these will be exactly true, but we can precisely quantify to what extent *Macbeth* appears to have been generated by a similar process to these two reference texts.

Of course, our model is just the distribution of characters, so is a fairly weak model of different styles.





Comparing Log-Likelihoods

```
[77]: macbeth_chars = numerify("data/macbeth.txt")
```

```
## A positive value indicates more likely to be "Like"
```

```
# Romeo and Juliet; negative is evidence
```

```
## in favour of Metamorphosis (i.e. written by Kafka)
```

```
print(log_likelihood(macbeth_chars, rj_pmf)-
```

```
      log_likelihood(macbeth_chars, metamorphosis_pmf))
```

```
113.97323849247186
```

```
[78]: ## We could repeat this with another text, in this case
```

```
## "The Trial", by Kafka
```

```
trial_chars = numerify("data/the_trial.txt")
```

```
## A positive value indicates more likely to be "Like" Romeo and Juliet
```

```
print(log_likelihood(trial_chars , rj_pmf)-
```

```
      log_likelihood(trial_chars, metamorphosis_pmf))
```

```
-928.3458629352972
```

Better text representation: bigrams

- We can look at these in the case of the character model of text we saw earlier. From the vector of character codes that make up "Metamorphosis" we can take every *pair of characters*, in the order they appear.
- That is, we consider two characters c_{i-1} and c_i at some index i . This is called a **bigram** model, and there are *unigram*, *trigram*, *n-gram* generalisations of the idea.
- A "gram" refers to a unit like a character or word, and we are discussing a **character bigram model**.

c_{i-1} c_i
Rom. Give me a torch. I am not for this ambling.
Being but heavy, I will bear the light.

Mer. Nay, gentle Romeo, we must have you dance.

Rom. Not I, believe me. You have dancing shoes
With nimble soles; I have a soul of lead
So stakes me to the ground I cannot move.



Joint probabilities

The Joint probability of two random variables X and Y is written:

$$P(X, Y)$$

And gives the probability that X and Y take specific values *simultaneously*

E.g.

$$P(X=x) \wedge P(Y=y)$$



Marginal probabilities

The **marginal probability** is the derivation of $P(X)$ from $P(X,Y)$ by integrating (summing) over all the possible outcomes of Y :

$$P(X) = \int_y P(X, Y) dy \text{ for a PDF.}$$

$$P(X) = \sum_y P(X, Y) \text{ for a PMF.}$$

This allows us to compute a distribution over one random variable from a joint distribution by summing over all the possible outcomes of the other variable involved.



Marginalisation

Marginalisation just means integration over one or more variables from a joint distribution: it *removes* those variables from the distribution.

- Two random variables are **independent** if they do not have any dependence on each other. If this is the case then the joint distribution is just the product of the individual distributions: $P(X,Y)=P(X)P(Y)$.
- This is **not true in the general case where the variables have dependence**.



Conditional probabilities

The **conditional probability** of a random variable X *given* a random variable Y is written as $P(X|Y)$ and can be computed as:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}.$$

This tells us how likely the outcomes of X are *if we already know* (or fix) the outcomes of Y .

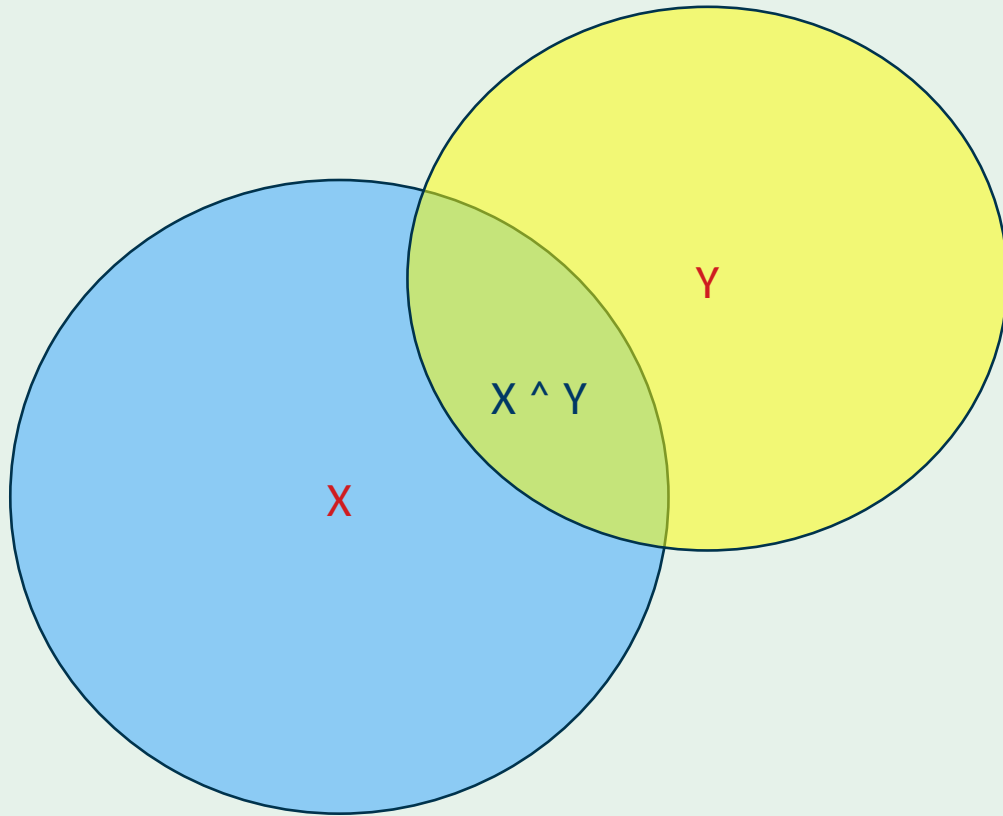
Read as "probability of X taking on the value x given that Y has taken on the value y ".

The conditional probability is $P(X|Y)=P(X)$ and $P(Y|X)=P(Y)$ if X and Y are independent.



Or visually

State space (all possible outcomes)



- What is $P(X)$?
- What is $P(X,Y)$?
- What is $P(X \vee Y)$?
- Let us marginalise $P(X,Y)$ over Y
- What is $P(X | Y)$?



Example: Back to the Scorpion

We have a grid of outcomes with 6 rows and 8 columns, for a total of 48 possible outcomes.

State space (all possible outcomes)



Example: Back to the Scorpion

- We have a grid of outcomes with 6 rows and 8 columns, for a total of 48 possible outcomes (from (1,1) to (6,8)).
- **Event X** (“south”) is true if the first coordinate is 4 or more.

Event X



Example: Back to the Scorpion

Event Y

- We have a grid of outcomes with 6 rows and 8 columns, for a total of 48 possible outcomes (from (1,1) to (6,8)).
- **Event X** (“south”) is true if the first coordinate is 4 or more.
- **Event Y** (“East”) is true if the second coordinate is 6 or more.



Example: Back to the Scorpion

- What is $P(X=\text{true})$?
- What is $P(Y=\text{true})$?
- What is $P(X=\text{true} \text{ and } Y=\text{true})$?
- What is $P(X=\text{true} \text{ or } Y=\text{true})$?
- What is $P(X,Y)$?
- Marginalise $P(X,Y)$ over Y .
- What is $P(X=\text{true} \mid Y=\text{true})$?



Example: Back to the Scorpion

- What is $P(X=\text{true})$?
 - $24/48 = 0.5$
- What is $P(Y=\text{true})$?
 - $18/48 = 3/8 = 0.375$
- What is $P(X=\text{true} \text{ and } Y=\text{true})$?
 - $9/48 = 3/16 = 0.1875$
- What is $P(X=\text{true} \text{ or } Y=\text{true})$?
 - Not independent, so
 $0.5+0.375-0.1875=0.6875$



Example: Back to the Scorpion

- What is $P(X,Y)$?
 - A matrix of probabilities for combinations of values for X and Y :

	$Y = \text{true}$	$Y = \text{false}$
$X = \text{true}$	9/48	15/48
$X = \text{false}$	9/48	15/48

Example: Back to the Scorpion

- What is $P(X,Y)$?
 - A matrix of probabilities for combinations of values for X and Y :

	$Y = \text{true}$	$Y = \text{false}$
$X = \text{true}$	$9/48$	$15/48$
$X = \text{false}$	$9/48$	$15/48$

- Marginalise $P(X,Y)$ over Y .
 - This gives $P(X)$

	$P(X)$
$X = \text{true}$	$9/48 + 15/48 = 24/48 = 0.5$
$X = \text{false}$	$9/48 + 15/48 = 24/48 = 0.5$



Example: Back to the Scorpion

- What is $P(X | Y)$?
 - $P(X | Y) = P(X,Y)/P(Y)$
 - We calculated $P(X,Y)$ before

$P(X,Y)$	$Y = \text{true}$	$Y = \text{false}$
$X = \text{true}$	$9/48 = 3/16$	$15/48 = 5/16$
$X = \text{false}$	$9/48 = 3/16$	$15/48 = 5/16$

- We marginalise $P(X,Y)$ over X

	$P(Y)$
$Y = \text{true}$	$(3+3)/16 = 3/8$
$Y = \text{false}$	$(5+5)/16 = 5/8$

- And finally calculate $P(X | Y)$

$P(X Y)$	$Y = \text{true}$	$Y = \text{false}$
$X = \text{true}$	$3/16 \times 8/3 = 0.5$	$5/16 \times 8/5 = 0.5$
$X = \text{false}$	$3/16 \times 8/3 = 0.5$	$5/16 \times 8/5 = 0.5$



Back to Bigrams



From the vector of character codes that make up "Metamorphosis" we can take every *pair of characters*, in order that they appear.

- The **joint distribution** of bigrams $P(c_i, c_{i-1})$ is given by the normalized count of each character pair.
- The **marginal distribution** $P(c_i)$ can be computed from $P(c_i, c_{i-1})$ by summing over every possible character c_{i-1} , and likewise to marginalise to find $P(c_{i-1})$
- The **conditional distribution** $P(c_i | c_{i-1})$ is given by the joint distribution, divided by the counts of $P(c_{i-1})$. It tells us how likely we are to observe a specific character c_i *given* that we have observed a character c_{i-1} just beforehand.



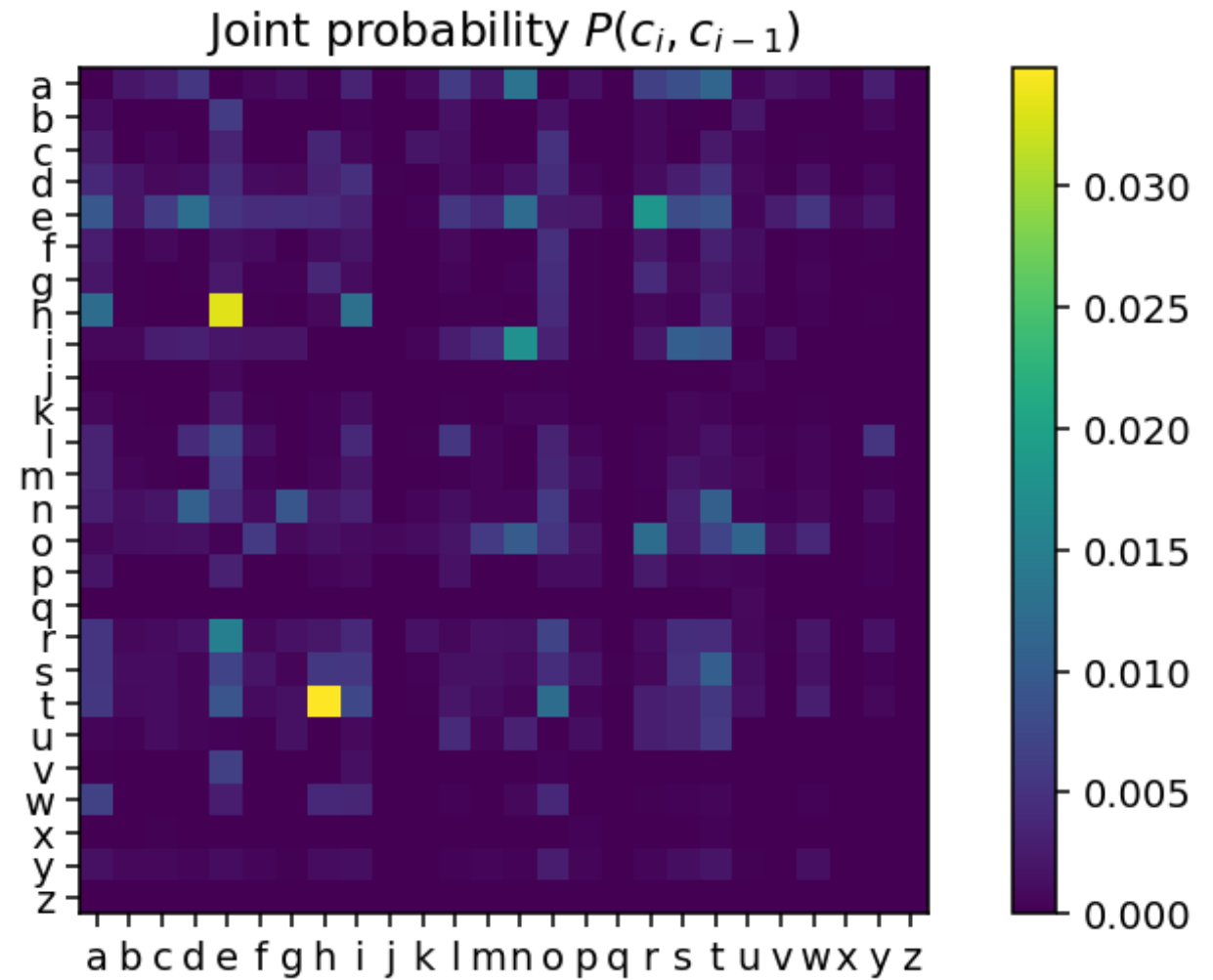
Bigrams: Joint probability

The joint probability for a bigram is $P(c_i, c_{i-1})$

```
[14]: meta_joint = np.zeros([len(alphabet), len(alphabet)])  
      for c_1, c_2 in zip(metamorphosis[1:], metamorphosis[:-1]):  
          meta_joint[c_2, c_1] += 1  
  
      # normalise by total number of pairs  
      meta_joint = meta_joint / np.sum(meta_joint)
```

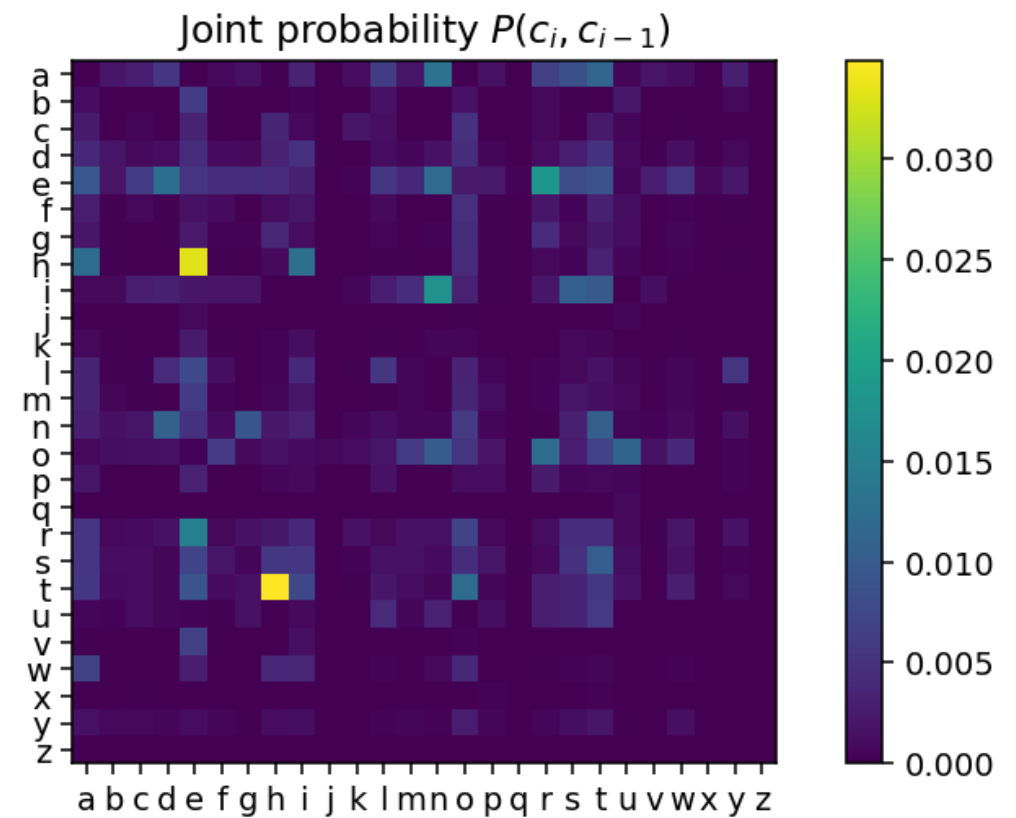
This tells us how likely each possible **pair** of characters is. Some combinations are much more likely than others.

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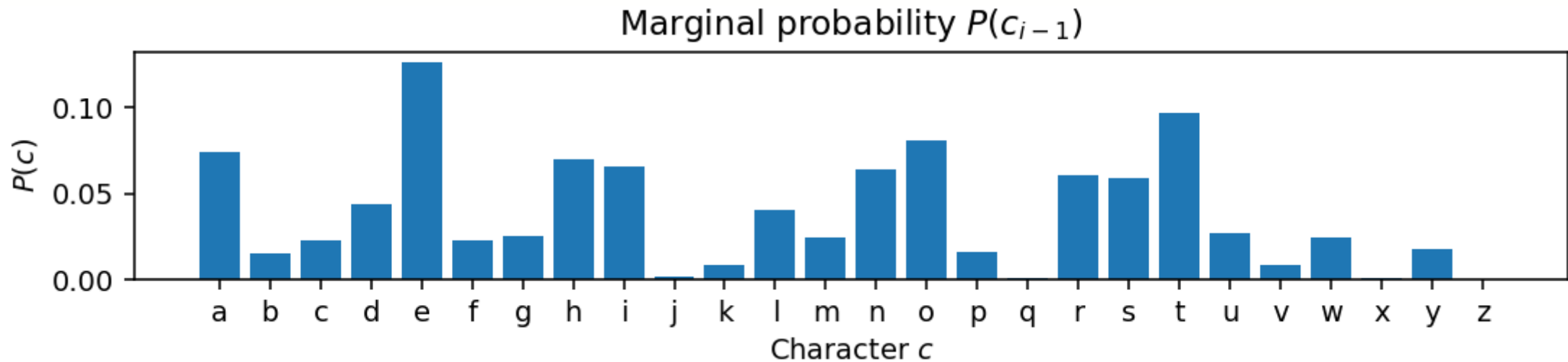
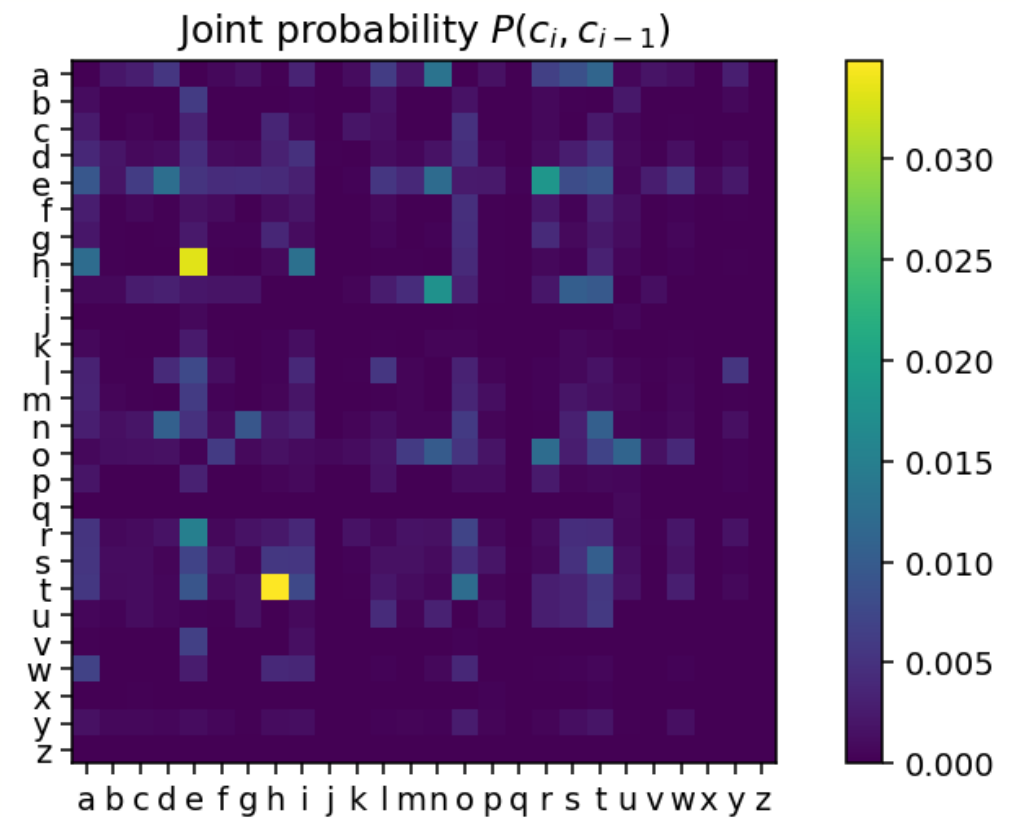
Bigrams: Marginalisation



```
[15]: # compute the marginal probabilities, by summing over the relevant axes  
      # it's this easy!  
      meta_c1 = np.sum(meta_joint, axis=0)  
      meta_c2 = np.sum(meta_joint, axis=1)
```

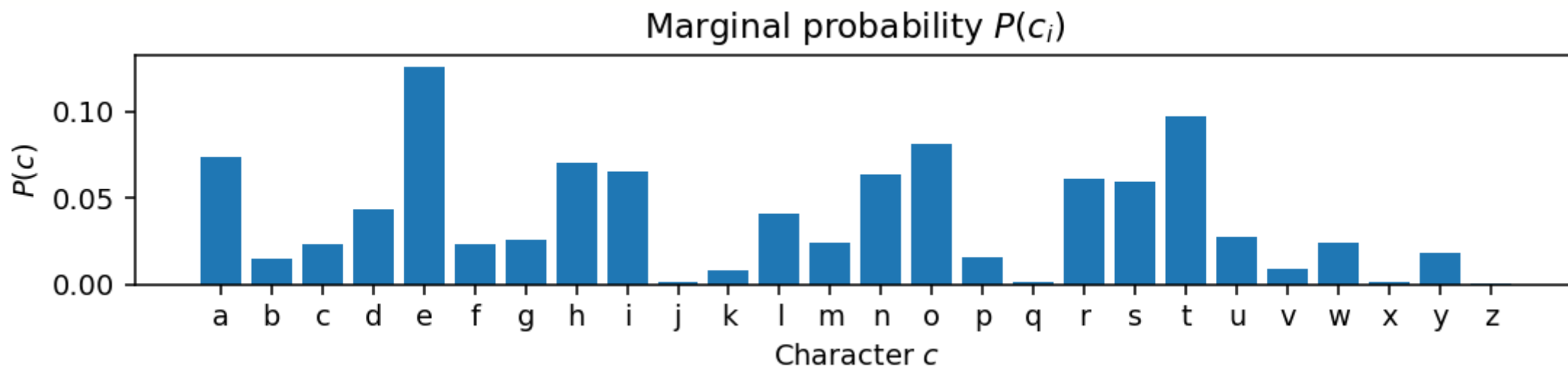
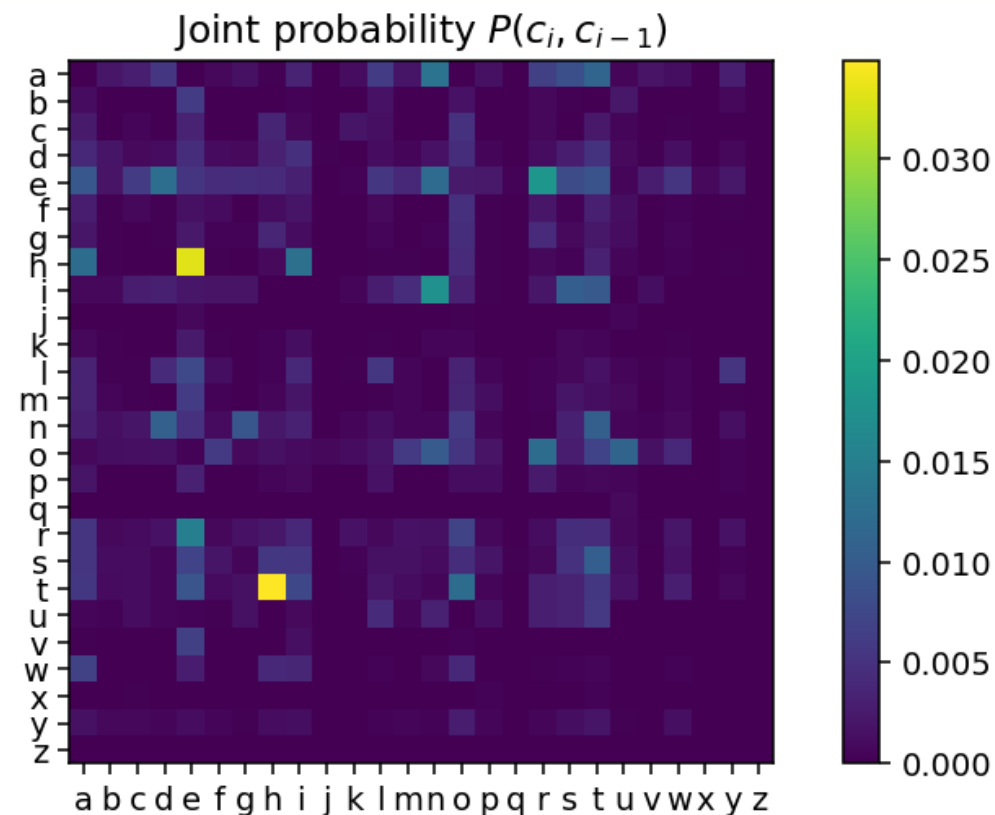


Bigrams: Marginalisation





Bigrams: Marginalisation





Bigrams: Conditional probabilities

The conditional probability $P(c_i|c_{i-1})$ is given by:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}.$$

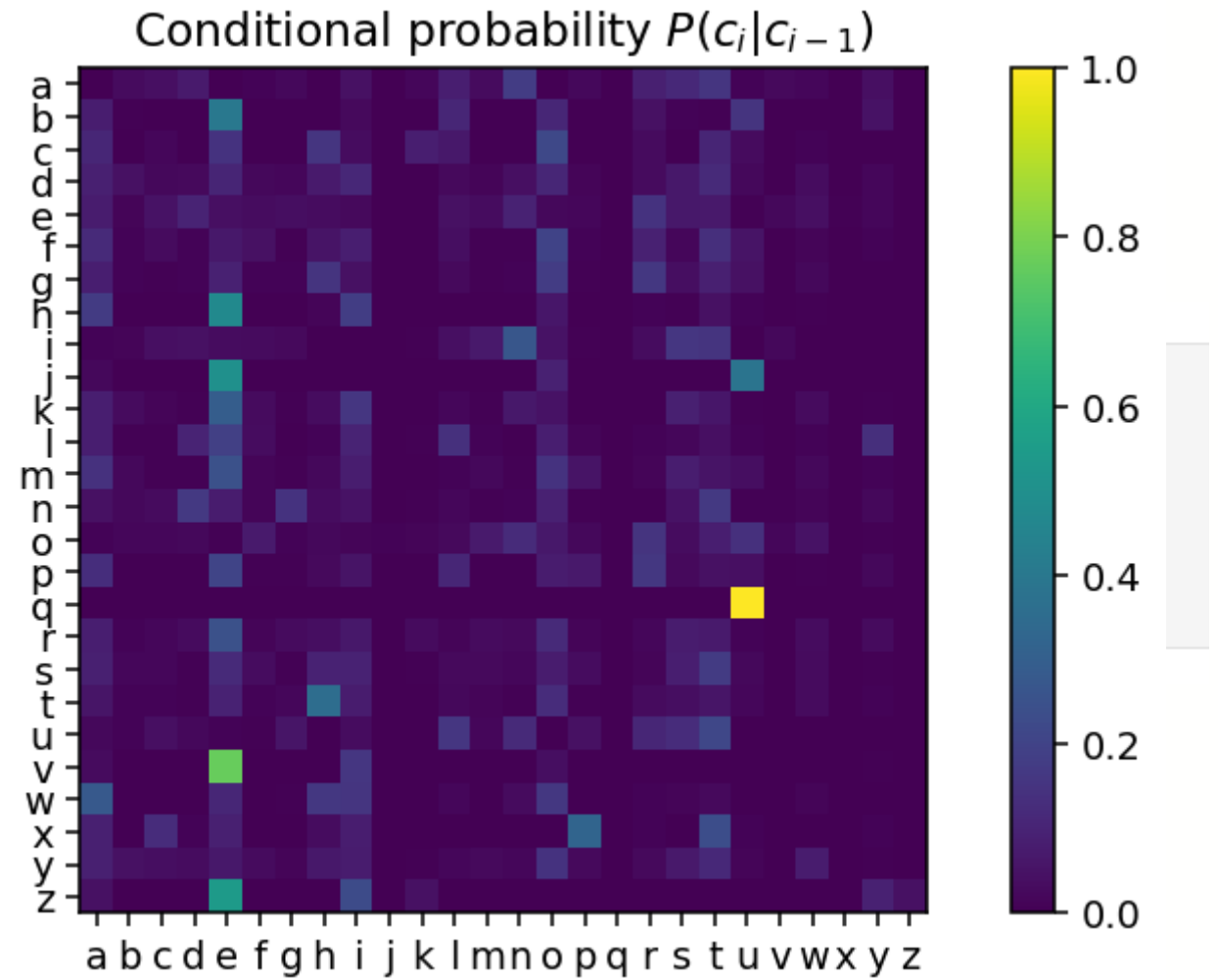
```
[23]: #  $p(c_2|c_1) = p(c_1, x_2) / p(c_1)$   
# have to transpose to normalise across columns, instead of rows  
meta_conditional = (meta_joint.T / meta_c1).T  
  
# plot it  
show_2d_histogram(meta_conditional)  
plt.gca().set_title("Conditional probability  $P(c_i|c_{i-1})$ ")  
plt.show()
```



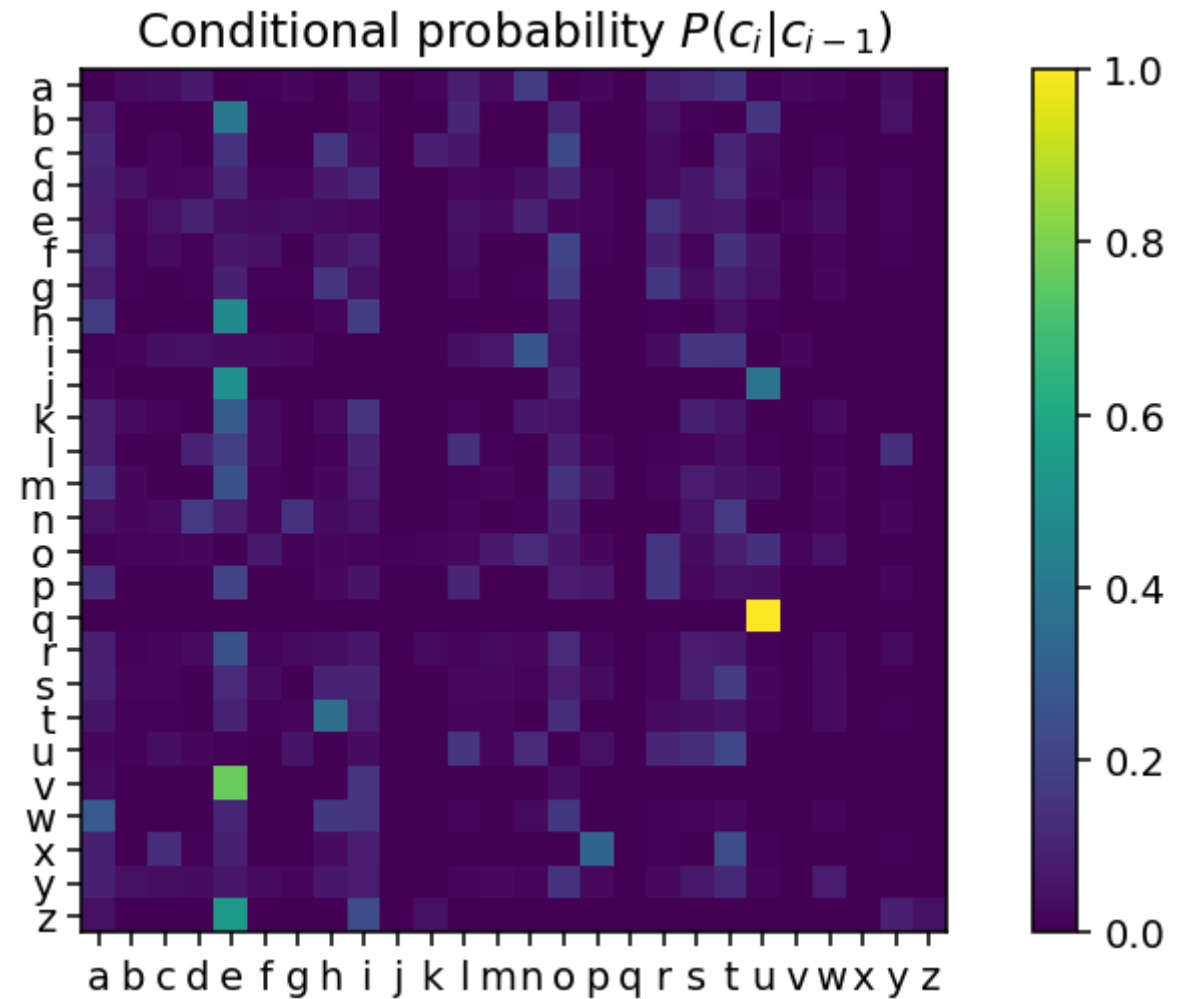
Bigrams: Conditional probabilities

The conditional probability $P(c_i|c_{i-1})$ is given by:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}.$$



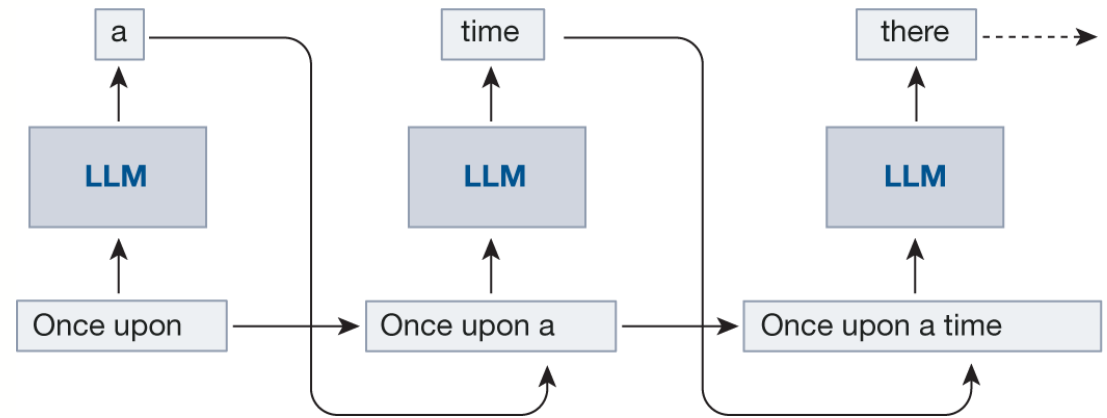
For example, look at row "q": the probability is concentrated on the entry in column "u" -- "u" always follows "q" in English.





Aside: Conditional probabilities and LLM

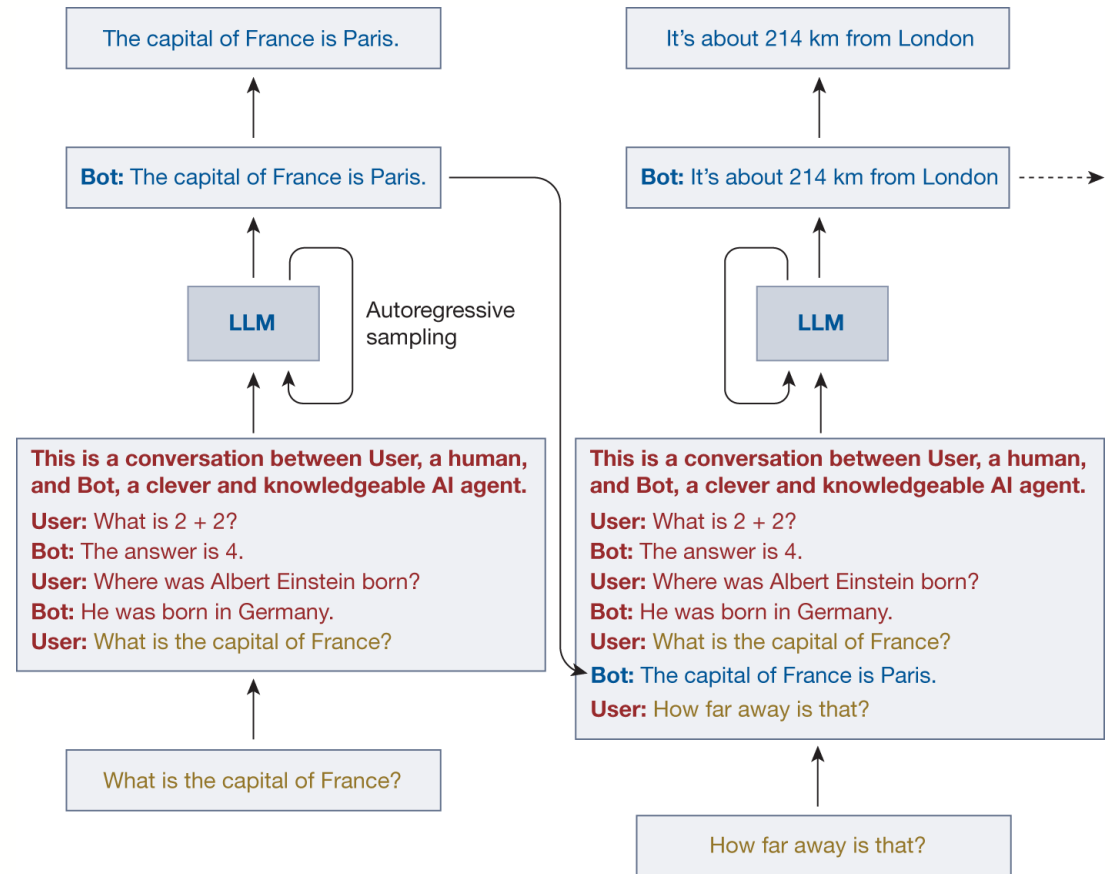
- In effect, this is the same strategy used by **Large Language Models** (eg, GPT) for generating text:
- They generate one word at a time by **sampling from the distribution of words conditioned to the previous text**
- This is including the user query and the previously generated words!
- This process is called *autoregressive*





Aside: Conditional probabilities and LLM

- In effect, this is the same strategy used by **Large Language Models** (eg, GPT) for generating text:
- They generate one word at a time by **sampling from the distribution of words conditioned to the previous text**
- This is including the user query and the previously generated words!
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Part 2 - Bayes' theorem



☠ Plague ☠



- A new test is developed that can detect *Plague X* with 95% accuracy.
- We'll assume that 95% accuracy means:
 - a 5% **false positive rate**, i.e. 5% of the time people who don't have the disease test positive
 - a 5% **false negative rate**, i.e. 5% of the time people have the disease test negative
- One in one hundred thousand (1:100000) people are known to have *Plague X*
 - we might know this because it's universally fatal within one year, and one in a hundred thousand people died of it last year, for example.
- You go for the test, and it comes back *positive* for *Plague X*.
- How likely are you to have *Plague X*?



<https://www.menti.com/alquw5aa3won>

Or code: 2987 1383



Bayes' rule

Inverting conditional distributions

We often want to know the probability of some event A given some other event B ; that is $P(A|B)$. But we are often in a situation where we can only compute $P(B|A)$.

This case is usually:

- we know how the mysterious entity behaves $P(B|A)$;
- we know what data we saw $P(B)$;
- we know what the mysterious entity is likely to be up to in general $P(A)$;
- and we want to work out what the mysterious entity is doing $P(A|B)$.

Bayes' rule

Inverting conditional distributions

We often want to know the probability of some event A given some other event B ; that is $P(A|B)$. But we are often in a situation where we can only compute $P(B|A)$.

In general $P(A|B) \neq P(B|A)$; and the two expressions can be completely different.

Typically, this type of problem occurs where we:

- want to know the probability of some event given some *evidence* (*how likely is it that I have a disease given that my blood test came back positive?*)
- but we only know the probability of observing evidence given the event (*if you have this disease, the blood test will come back positive 95% of the time*).



Bayes' rule

Inverting conditional distributions

We often want to know the probability of some event A given some other event B ; that is $P(A|B)$. But we are often in a situation where we can only compute $P(B|A)$.

Bayes' rule gives the correct way to invert the probability distribution:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

This follows directly from the axioms of probability. Bayes' Rule is a very important rule, and has some surprising consequences.



Bayes' rule Nomenclature

- $P(A|B)$ is called the **posterior** -- what we want to know, or will know after the computation
- $P(B|A)$ is called the **likelihood** -- how likely the event A is to produce the evidence we see
- $P(A)$ is the **prior** -- how likely the event A is regardless of evidence
- $P(B)$ is the **evidence** -- how likely the evidence B is regardless of the event.

Bayes' rule gives a consistent rule to take some prior belief and combine it with observed data to estimate a new distribution which combines them.



Bayes' rule Nomenclature

We often phrase this as some **hypothesis** H we want to know, given some **data** D we observe, and we write Bayes' Rule as:

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

H and D are random variables in this expression.

(the probability of the hypothesis given the data) is equal to (the probability of the data given the hypothesis) times (the probability of the hypothesis) divided by (the probability of the data).

In other words, if we want to work out how likely a hypothesis is to be true given observations, but we only know how likely we are to have seen those observations if that hypothesis *was* true, we can use Bayes' rule to solve the problem.



Bayes' rule

Marginalising the evidence

We can say that the posterior probability is *proportional* to the product of the prior and the likelihood. But to evaluate its value, we need to compute $P(D)$, **the evidence**.

It is difficult to see what this represents at first. But one way to think of it is as the result of marginalising the $P(D)$ from the joint distribution $P(H,D)$; that is integrating $P(H,D)$ over every possible outcome of H , for each possible D .

Because probabilities must add up to 1, we can write $P(D)$ as:

$$P(D) = \sum_i P(D|H_i)P(H_i)$$

for a set of discrete outcomes H_i or

$$P(D) = \int_H P(D|H)P(H)dH$$

for a continuous distribution of outcomes.

This trick is essential in understanding Bayes' Rule!



Bayes' Rule Binary cases

In general this can be difficult to compute. For binary simple cases where there are only two possible outcomes (H can only be 0 or 1), Bayes' rule can be written as:

$$P(H = 1|D) = \frac{P(D|H = 1)P(H = 1)}{P(D|H = 1)P(H = 1) + P(D|H = 0)P(H = 0)},$$

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$



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Bayes' rule Back to Plague X



Bayes Rule for *Plague X*

$$P(\text{Plague}|\text{Test}) = [P(\text{Test}|\text{Plague}) P(\text{Plague})] / P(\text{Test})$$

integrate over the two possible
#states of Plague

$$P(\text{Test}) = [P(\text{Test}|\text{not Plague})P(\text{not Plague}) + \\ P(\text{Test}|\text{Plague})P(\text{Plague})]$$



Bayes' rule

Answering the Plague X question



```
[18]: # one in a ten thousand have the plague
      P_Plague = 1.0 / 100000

      # if you have the plague, test is positive with 5% error
      P_Test_Plague = 1-0.05
      # If you don't have the plague, the test is positive 5% of the time
      P_Test_not_Plague = 0.05

      # chance you don't have the plague, before we see the test
      P_not_Plague = 1-P_Plague

      # integrate over evidence; only two possibilities here
      P_Test = (P_Test_Plague * P_Plague +
                P_Test_not_Plague * P_not_Plague)

      # what is the probability you have the plague,
      # given this 95% accurate test?
      # (hint: it is not 95%!)
      P_Plague_Test = P_Test_Plague * P_Plague / P_Test

      print("After testing positive, you have a 1:%.0f chance of having the plague" % ((1-P_Plague_Test)/P_Plague_Test))

      After testing positive, you have a 1:5263 chance of having the plague
```

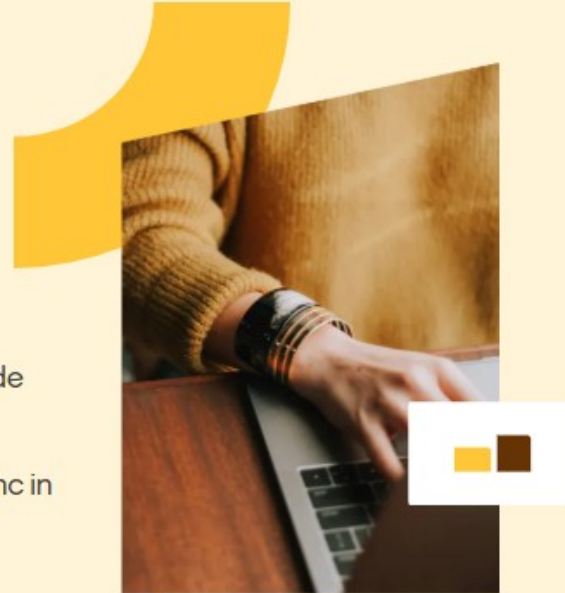


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Bayes' rule

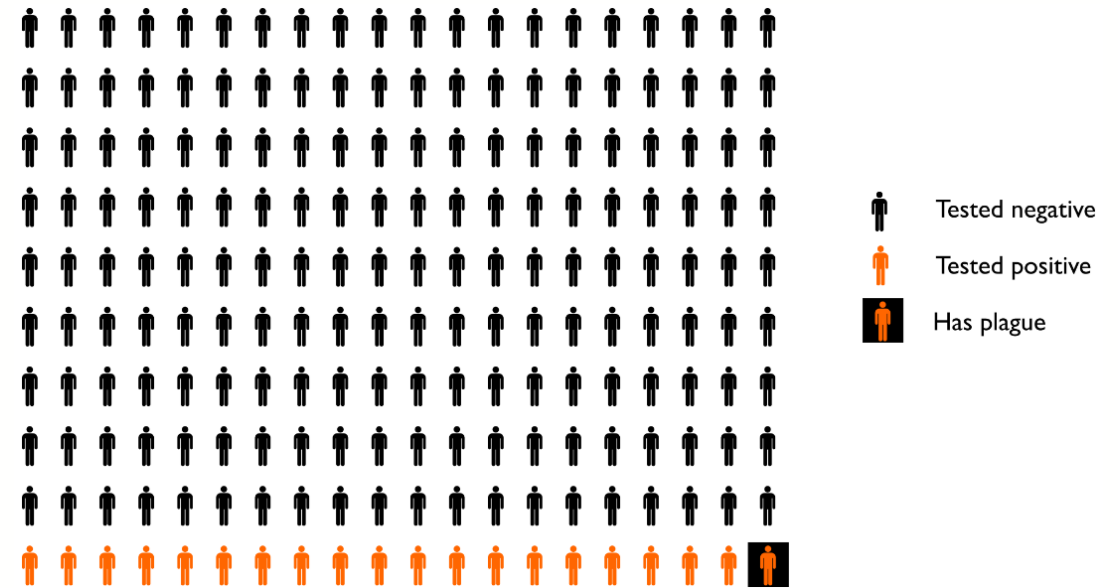
Natural frequency

There is an approach to explaining problems like this which makes it much less likely to make poor judgements.

Natural frequency explanations involve imagining concrete populations of a fixed size (*10000 people in a room, for example*), and considering the proportions of the populations as *counts* (*how many people in the room have the plague?*).

We can use this to visualise the problem above and explain the apparent paradox.

The graph on the right shows the case $P(\text{plague})=0.005$ (1 in 200), again with a 5% accurate test.





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Part 3 - Entropy

entropy



Entropy

A key property of a probability distribution is the **entropy**. Intuitively, this is a measure of the "surprise" an observer would have when observing draws from the distribution, or alternatively, the (log) measure of a number of distinct "states" a distribution could represent.

A flat, uniform distribution is very "surprising" because the values are very hard to predict; a narrow, peaked distribution is unsurprising because the values are always very similar.



Entropy

This is a precise quantification -- it gives the *information* in a distribution. The units of information are normally bits; where 1 bit of information tells you the answer to exactly one yes or no question.

The entropy tells you exactly how many of bits are needed (at minimum) to communicate a value from a distribution to an observer *who knows the distribution already*.

Alternatively, you can see the number of distinct states the distribution describes as $p=2^{H(X)}$ -- this value is called the **perplexity**, and it can be fractional.



Entropy

The entropy of a (discrete) distribution of a random variable X can be computed as:

$$H(X) = \sum_{\mathbf{x}} -P(X = \mathbf{x}) \log_2(P(X = \mathbf{x}))$$

This is just the expected value of the log-probability of a random variable (the "average" log-probability).



Example: Tossing coins

Consider a coin toss: this is sampling from a discrete random variable that can take on two states, **heads** and **tails**.

If we call our two possible states 0 (heads) and 1 (tails), we can characterise this with a single parameter q , where $P(X=0)=q$ and $P(X=1)=(1-q)$

(this follows from the fact that as $P(X=0)+P(X=1)$ must equal 1 -- the coin must land on one side or the other. We ignore edge landings!).

Example: Tossing coins

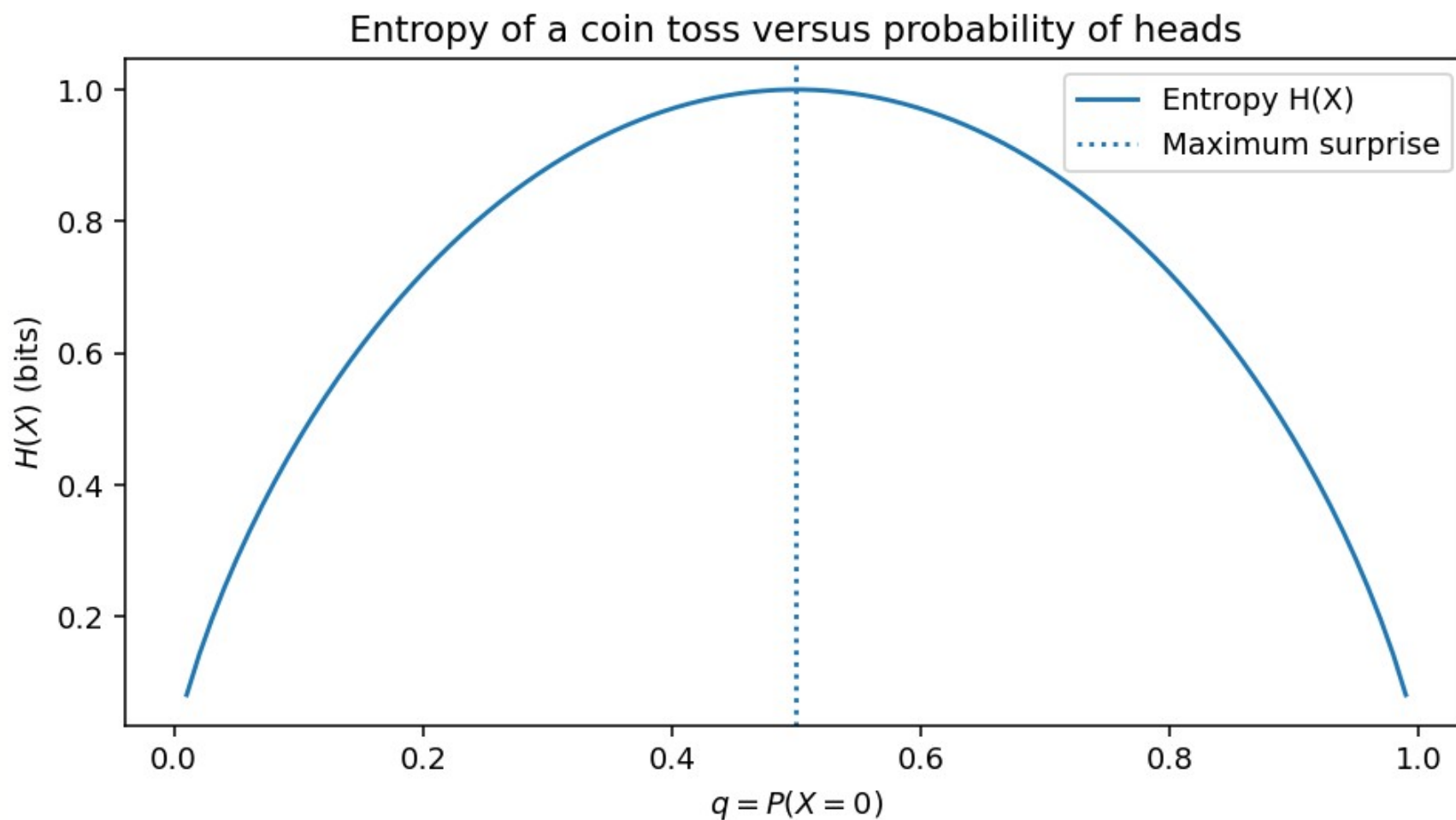
- If this process is very biased and heads are much more likely than tails ($q \ll 0.5$), an observer will be unsurprised most of the time because predicting heads will be a good bet.
- If the process is unbiased ($q=0.5$), an observer will have no way to predict if a head or tail is more likely.

We can write the entropy of this distribution:

$$\begin{aligned} H(X) &= P(X=0) \log_2 P(X=0) + P(X=1) \log_2 P(X=1) \\ &= -q \log_2 q - (1-q) \log_2 (1-q) \end{aligned}$$



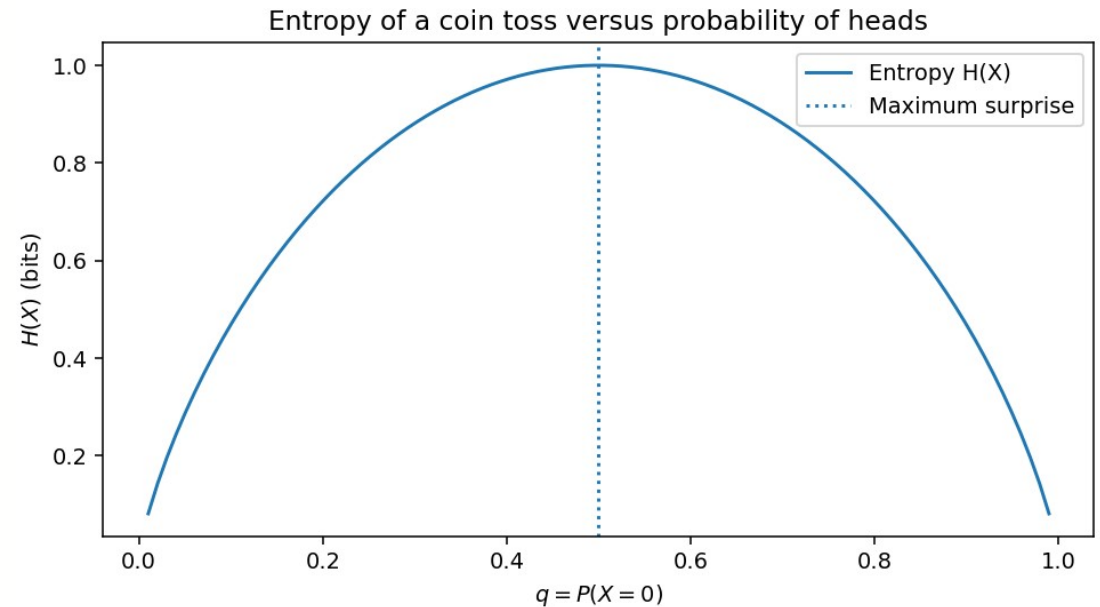
Entropy of a coin toss





Entropy of a coin toss

- This corresponds with our intuition:
an unbiased coin toss is the most surprising distribution.
- Entropy is a critically important concept in **information theory** which relates probability to the problems of communication.
- We can measure how well Bayesian updating is going by measuring the *drop in entropy of the posterior belief*; as we add evidence, we should become less surprised about the possible outcomes.

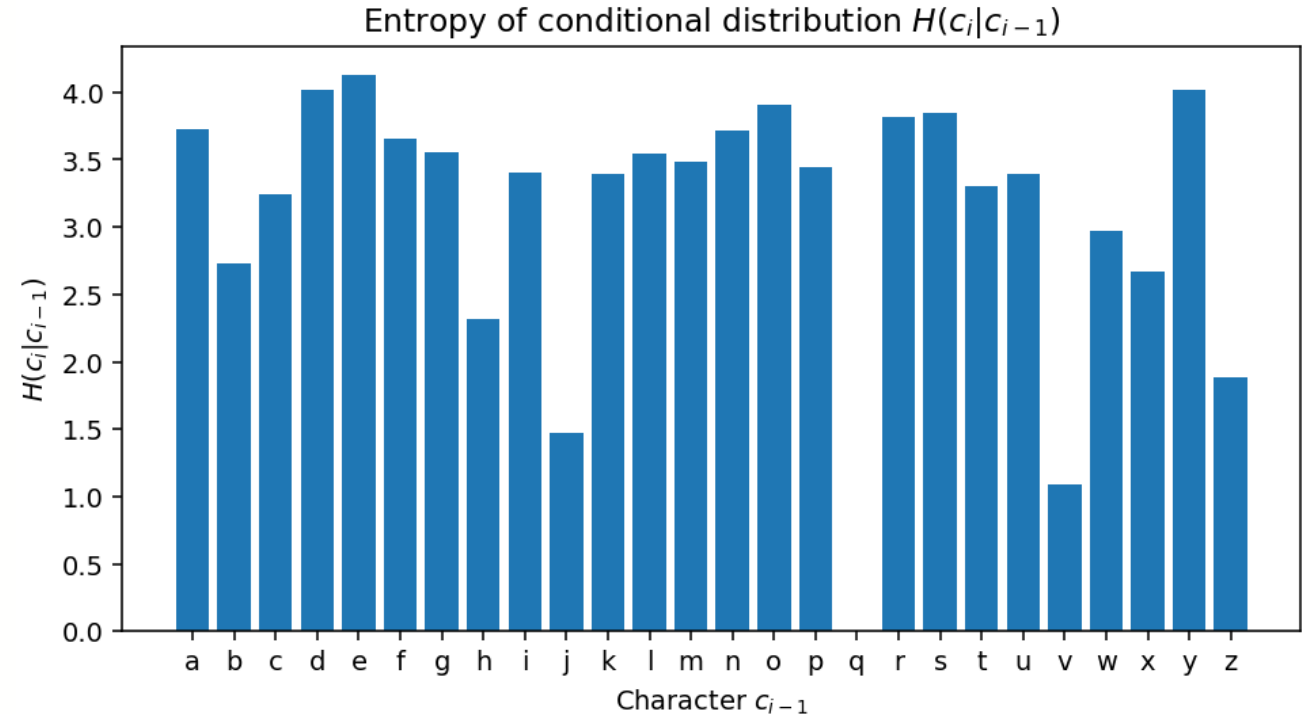




Example 2: Back to our bigrams

In our letter-based bigram example, we can compute the entropy of each conditional distribution $P(c_i|c_{i-1})$.

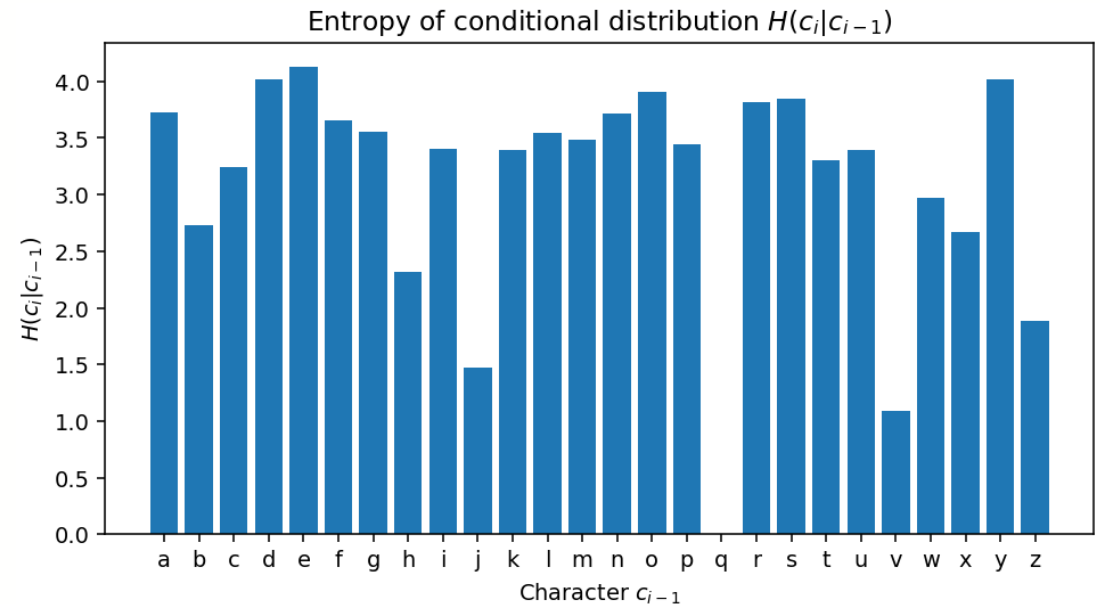
This will tell us how surprised we will be by the characters that might follow a character c_i .





Interpreting entropy

- We can see that observing a "q" means the next character **isn't surprising at all**: we know for **sure** that it will be a "u", and thus the entropy is 0. There is only one configuration of the world for the next character (under this model) if we have just seen a "u".
- Likewise, seeing a space character gives us **very little information**, and **the next character could be anything**: the character after a space will surprise us -- there are lots of configurations of the world that might follow a space. That surprise is as much surprise as we would get by tossing about 4 coins, or an entropy of about 4 bits.





Summary

We can represent probabilities between multiple random variables using **joint**, **marginal** and **conditional** probabilities

Bayes' rule allows to invert a conditional probabilities, and crucially to **update our belief** (posterior) when new evidence is available.

Entropy allows to model **uncertainty** and will be useful to assess how effective a Bayesian update was at increasing our knowledge.



University
of Glasgow

Thank you

**Contact:
Nicolas.Pugeault@Glasgow.ac.uk**